LAB 4 – VIBECODING CHAT HISTORY

Relevant ChatGPT interaction record for the Helicopter game implementation

Objective

Use Vibe Coding with an LLM to fix the broken Helicopter game and implement the assigned features. This record contains only the relevant coding interaction, implementation summary, and verification for the four tasks.

Task 1 – Fix Helicopter Movement & Boundary

Prompt / request

> Fix the helicopter movement so upward velocity does not keep accumulating and add a maximum speed limit. Also fix the bottom boundary so the helicopter cannot leave the screen when Down is held.

LLM-assisted implementation

ChatGPT-assisted implementation: Updated vertical movement and boundary handling so the helicopter stays within the playable screen. Separate commit: “Fix helicopter movement and boundaries”.

Verification

Test result: Movement and screen boundaries worked correctly.

Task 2 – Obstacle Collision & Game Over

Prompt / request

> Add obstacle collision and game-over behavior. Touching the top or bottom wall should end the game, while passing through the safe gap should not.

LLM-assisted implementation

ChatGPT-assisted implementation: Added obstacle/wall collision detection and game-over handling. Separate commit: “Add obstacle collision and game over”.

Verification

Test result: Collision and game-over behavior worked correctly.

Task 3 – Distance-Based Scoring

Prompt / request

> Add distance-based scoring. Track distance travelled, display the score during the game, show final distance on game over, and reset the distance when a new game starts.

LLM-assisted implementation

ChatGPT-assisted implementation: Added distance tracking, score display, final distance, and reset behavior. Separate commit: “Add distance based scoring”.

Verification

Test result: Score increased during gameplay and reset/final-distance behavior worked correctly.

Task 4 – Shield Mechanic

Prompt / request

> Add a shield that can be activated, visibly indicates when active, protects the helicopter from one obstacle collision, disappears after one hit, and leaves the helicopter vulnerable until reactivated.

LLM-assisted implementation

ChatGPT-assisted implementation: Added the shield mechanic in the game engine, renderer, and main controls. Shield activation uses S. It absorbs exactly one obstacle collision and then disappears. Separate commit: “Add shield

mechanic”.

Verification

Test result: Shield activation, one-hit protection, disappearance after impact, and subsequent vulnerability all worked correctly.

Final Verification

All four assigned tasks were implemented and tested successfully. Each task was kept as a separate Git commit, and the completed project was pushed to the personal GitHub repository.

Deliverable item	Status
Movement & boundary fix	Completed / tested
Obstacle collision & game over	Completed / tested
Distance-based scoring	Completed / tested
Shield mechanic	Completed / tested
Separate commits	Completed
Personal GitHub push	Completed

Note: This document intentionally excludes unrelated environment setup, installation, authentication, and troubleshooting steps and focuses on the Vibe Coding work relevant to Lab 4.